Outlook

Re: HOBBYWING New customer message on April 6, 2025 at 1:18 pm

From Xinya Lin <xin.y.lin@mail.mcgill.ca>
Date Mon 4/7/2025 11:24 PM
To Malo Salmon <malo.salmon@mail.mcgill.ca>

He probably doesn't know much either. You can either try to contact the Chinese support or just let it be.
Thanks a lot though

From: Malo Salmon <malo.salmon@mail.mcgill.ca>
Sent: Monday, April 7, 2025 10:10:58 PM
To: Xinya Lin <xin.y.lin@mail.mcgill.ca>
Subject: Fw: HOBBYWING New customer message on April 6, 2025 at 1:18 pm

This guy is useless. Clearly doesn't want to answer the question.

From: HW Support MC (HOBBYWING Customer Care) <onlineorders@hobbywingdirect.com>
Sent: Monday, April 7, 2025 9:56 PM
To: Malo Salmon <malo.salmon@mail.mcgill.ca>
Subject: Re: HOBBYWING New customer message on April 6, 2025 at 1:18 pm

You don't often get email from onlineorders@hobbywingdirect.com. Learn why this is important

Hello Malo,

There is no direct correlation between the load and amp draw, there are many other factors that will play in to the current the system will draw.

Best Regards,
Sam

On April 8, 2025 at 1:50:57 AM UTC, Malo Salmon malo.salmon@mail.mcgill.ca wrote:

> Hello,
>
> Would it be possible to know the mathematical relationship between the curent draw and the load?
>
> Thank you,
>
> Malo
>
> ---
>
> **From:** HW Support MC (HOBBYWING Customer Care) <onlineorders@hobbywingdirect.com>
> **Sent:** Monday, April 7, 2025 12:13 PM

To: Malo Salmon <malo.salmon@mail.mcgill.ca>
Subject: Re: HOBBYWING New customer message on April 6, 2025 at 1:18 pm

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Hello Malo,

The current draw on the system will depend on the load.

The FOC drive of the system will alter the torque from the motor depending on the load applied.

Larger load = more amp draw.

Please note that the 40A is the max continuous draw that the motor will have.

Best Regards,
Sam

On April 6, 2025 at 8:18:51 PM UTC, Malo Salmon malo.salmon@mail.mcgill.ca wrote:

You received a new message from your online store's contact form.

Country Code:
CA
Your Full Name:
Malo Salmon
Email:
malo.salmon@mail.mcgill.ca
Message:
Hello,

I am currently working on a project in which I am using the Hobbywing Quicrun Fusion SE (2in1 FOC System) electric motor. This project requires us to calibrate the rotation speed of the motor very precisely. The output RPM is a function of both the voltage supplied to the motor and the PWM (pulse width modulation/duty cycle) given. i.e.

Output RPM = f(V,PWM)

V : Input Voltage (V)

PWM : Input duty cycle (%)

This means that to output a given speed, we need to know what the given input voltage and the given input PWM are. The duty cycle input can be precisely given by our microcontroller and is therefore not a problem. However, we would like the voltage input to the motor to be constant, such that the output RPM of the motor is only a function of the duty cycle. For this, we need to design a powerboard that regulates the voltage of our battery such that the voltage applied at the motor is constant (7.4V).

To design this powerboard, we need to know the nominal current draw of the motor so that we can choose the right components. While the datasheet of the motor indicates that there is a max current draw of 40A, and that the BEC provides 4A continuous current, we're not quite sure about how to interpret this data and include it in our power board design.

From our testing, we noticed that the current draw between the battery and the motor is quite small (~0.1 to 1A). Can we take this as the nominal current draw for our power board design? If this is incorrect, or if there is anything that we have omitted, would you please let us know so that we can design our power distribution board accordingly?

Thank you for your time,

Malo

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